The wiki answers most of the open questions, and two things in it change how your team should play this.

Two rules that matter most for you

First, a low profit score doesn't knock you out. The organizers say outright that they reserve finalist spots for the best abstracts. That means the write-up, which is the part you're best placed to own, is a real way into the top 8 on its own, not just a side deliverable.

Second, breadth is graded as "do you trade all the products, and with different strategies?" A team that only trades three products loses points even if those three make money. Your team needs at least a basic strategy for every product, and the strategies need to differ.

The abstract itself has a 5-page cap, but they say one clear page beats five pages of filler. Put your team number and members at the top.

The 20 products, grouped by what they actually are

Group 1: six standalone products. These are AMETHYST, TOFFEE_CAKE, PAPER_LANTERNS, TELLURIUM, JELLY and PEANUT_BUTTER, each with a limit of 40 or 50 units. Each is its own little market with its own personality. Some will probably sit near a fixed price, some will drift, some will be jumpy. You can't tell which until someone charts the data. The default play is the market-making idea from before: quote just inside the spread around your fair-value estimate. The skill is estimating fair value differently depending on how each product behaves.

Group 2: the basket. FRUIT_BASKET is made of DRAGONFRUIT, GRANADILLA and MANGOSTEEN. You'll understand this one instantly, because it's ETF-vs-NAV arbitrage. The basket should be worth roughly the sum of its parts. If it trades above that, you sell the basket and buy the fruits; if it trades below, you do the reverse, and you profit when the gap closes.

The catch is that the wiki doesn't say how many of each fruit are in a basket. Working out those weights from the price data is a real task, and a nice "here's how we figured it out" section for the abstract. Also note that every product in this group has a limit of 40, which caps how big the hedge can get.

Group 3: options. LYCHEE is the underlying asset, with a limit of 200. The LYCHEE_VOUCHERS are call options on it at ten different strike prices, 50 units each.

A call with strike 7,500 is the right to buy LYCHEE at 7,500. So if LYCHEE is at 8,000, that right is worth at least 500, plus extra for the chance LYCHEE rises further before expiry. "No execution or settlement" means you can never actually exercise them. You just buy and sell the vouchers like any other product.

They expire 8 days after day −2. That makes 6 days left on day 0 and 5 days left on day 1, the scoring day. The standard approach is to price each voucher with Black–Scholes and back out implied volatility, which is the market's guess at how much LYCHEE will move. Then compare that across the ten strikes and trade the ones that look mispriced relative to the others. LYCHEE's big 200 limit is there so you can hedge, meaning offset the option risk by holding the underlying.

This is the most finance-flavoured part of the competition. If you've seen options at all, it's where you can contribute the most thinking.

Group 4: five pearls. WHITE, BLACK, GOLDEN, PINK and BLUE, limit 20 each, with no notes at all. Five products with matching limits and colour names almost certainly have some hidden relationship. Maybe they move together, maybe one leads the others, maybe one is a combination of the rest. Finding that relationship is a pure data-detective job.

How the backtester fills your orders, and why it flatters you

If you post an order to buy at 100, the backtester counts it as filled whenever a real historical trade happened at 100 or lower. That's generous. In a real market you'd be competing with other people for those fills.

So your backtest profit will likely overstate how a market-making strategy really performs. Expect it to come out lower on the scoring day, and don't over-celebrate backtest numbers. That's also a good "limitations" paragraph for the abstract, because judges like teams who know where their own results might be wrong.

What this means for you

You don't have to code to carry real weight. The organizers even mention using Codex to write the code, so AI-assisted coding is expected. Three things fit you:

- **The abstract.** Own it.
- **The basket and options logic.** Define the reasoning (how fair value is calculated, when to trade, how to hedge) and hand that to a teammate to implement.
- **The data analysis.** If you upload `prices.csv` and `trades.csv` here, I can chart every product, sort them by behaviour, estimate the basket weights, and back out the voucher volatilities. That gives your team a strategy map and gives you ready-made charts for the abstract.

One small thing for your coders: the wiki's NVIDIA example uses a variable called `product` that's never defined, so it would crash if copied as-is. The example from the slides had a similar `json.dump` vs `json.dumps` mistake.

The CSVs are in the Google Drive folder the wiki mentions. Can you download `prices.csv` and `trades.csv` for days −2, −1 and 0 and upload them here?

Got them: price and trade data for all three days, plus `datamodel.py` and the three example AMETHYST traders. I ran through every product. Here's what the data says, in plain terms.

Two practical things first. The voucher symbols in the data are `VOUCHER_6800`, `VOUCHER_7000` and so on, not `LYCHEE_VOUCHERS_6800` as the wiki says, so the code must use the data's names. And the days run continuously: each day opens where the last one closed. So the scoring day (day 1) will most likely pick up from day 0's closing prices, with 5 days left on the vouchers.

The standalone products: three are easy, one is a trap

AMETHYST is pinned at 10,000 and never moves. Buy below 10,000, sell above. The organizers' example trader already does this.

TOFFEE_CAKE is the best of this group. It hovers around 5,000 and snaps back fast: half of any move away from 5,000 disappears within about 70 ticks. Its spread is 15 coins, the widest of any product. You can quote well inside that gap around 5,000 and earn a big margin on every fill.

PAPER_LANTERNS looks jumpy tick to tick, but most of that is noise. The price flips back and forth around a value that drifts slowly. The fix is to use a smoothed average of recent prices as fair value instead of the current mid.

JELLY and PEANUT_BUTTER wander randomly with no pattern, so there's no edge in predicting them. Your RCFA bot's approach fits here: quote both sides, skew to stay near zero inventory, earn the spread.

TELLURIUM is the trap. It has by far the most trading of any regular product, with roughly 3,700 trades versus 100 to 300 for most others. But those trades are "informed": when someone buys at the ask, the price tends to rise over the next 20 ticks. A naive market maker on TELLURIUM keeps selling to people who know the price is about to go up. That's called adverse selection, and it's how market makers lose money. So TELLURIUM needs either wider quotes or a strategy that follows the trade flow instead of fading it.

The basket: the arbitrage is real but loose

The weights are 2 DRAGONFRUIT + 3 GRANADILLA + 1 MANGOSTEEN. Regressions at several time horizons all land on roughly 2 / 3 / 1, and tick by tick the basket moves with its parts (correlation 0.79).

The bad news is that the premium, meaning the basket price minus the value of its parts, isn't anchored. It averaged −187 on day −2, +516 on day −1 and −149 on day 0, and swings by hundreds within a day. That's not the clean "gap always closes" setup you'd hope for. The realistic version is to trade only big deviations from a rolling average, not from a fixed number.

The position limits also bite. Hedging one basket takes 3 GRANADILLA, so a 40-unit limit lets you fully hedge only 13 baskets. Treat this as a moderate opportunity, not the centrepiece.

The pearls: the best find in the data

The five pearl prices always add up to almost exactly 50,000. The sum's standard deviation is just 5 coins over three days. It's a hidden conservation rule: when one pearl goes up, the others go down by the same total.

There are three sub-patterns:

- **PINK_PEARLS does the correcting.** When the five prices add up to more or less than 50,000, PINK moves to close about two-thirds of that gap within 50 ticks. So PINK's real fair value is 50,000 minus the other four. That's a genuine predictive signal for trading PINK.
- **GOLDEN_PEARLS is a staircase.** It sits flat at 8,000, then steps down to 7,900, then 7,800, then 7,750, with brief slides between levels. Market-make it like AMETHYST around whichever level it's on, and stop quoting while it's stepping.
- **WHITE and BLACK move opposite each other.** When one ticks up, the other usually ticks down (correlation −0.68).

This group is where you show depth. Finding a hidden rule from data is exactly what judges want to see explained.

The options: the most money, and where your finance skills fit

The vouchers trade heavily, around 1,700 trades each, so there are lots of fill opportunities. Two findings:

1. **The options look expensive.** The prices imply LYCHEE will move about 1.9% to 2.8% per day. It actually moved 1.8% per day, very consistently, on all three days. So in-sample, the options are overpriced, especially on day 0. Selling them and hedging with LYCHEE would have made money. The risk is that if LYCHEE suddenly gets jumpy on the scoring day, short options lose badly. That's a trade-off to discuss in the abstract.
2. **Individual strikes drift off the smile and snap back.** If you plot implied volatility against strike, you get a smile shape, with the wings higher than the middle. Individual vouchers stray from that curve by about ±5 coins and snap back within a couple hundred ticks. Their spread is only 3 coins, so that's a tradable relative-value signal: buy the voucher that's cheap versus the curve, sell the one that's rich.

Where the profit will come from

Because of the backtester's fill rule, your passive orders only fill when real trades happen. Most regular products only see 50 to 100 trades a day, so profit will be concentrated where the volume is: the vouchers and TELLURIUM, plus TOFFEE_CAKE's wide spread and the pearl signal. That should guide where your team spends its limited time.

Next step: I can build a local version of their backtester (same fill rule) plus a full starter `Trader` that trades all 26 symbols with the strategies above. Then your team iterates on settings instead of starting from a blank file. Or I can make the charts for the abstract first. Which do you want?